Solve for PV by substituting t = 17 into the prior calculation using Equation 5:

$$PV = \text{INR}100 / (1 + 0.067)^{17} = \text{INR}33.21.$$

The INR5.88 price increase with a constant r represents implied interest earned over the three years. If the interest rate is positive, the PV generally rises (or accretes) over time to reach FV as time passes and t approaches zero.

3. Prices also change as interest rates change. Suppose after purchase at t = 0 we observe an immediate drop in the bond price to INR22.68224 per INR100 of principal. What is the implied interest rate on the discount bond?

Solution:

7.70 percent
Here we may solve for r in Equation 5 as follows:

$$\text{INR}22.68224 = \text{INR}100 / (1 + r)^{20}.$$

Rearranging Equation 5, we get:

$$r = 7.70 \text{ percent} = (100/22.68224)^{1/20} - 1.$$

Alternatively, we may use the Microsoft Excel or Google Sheets RATE function:
`= RATE (nper, pmt, PV, FV, type, guess)` using the same arguments as above, with *guess* as an optional estimate argument, which must be between 0 and 1 and defaults to 0.1, as follows:

7.70 percent `= RATE (20,0,-22.68224,100,0,0.1)`.

This shows that a 100 basis point (1.00 percent) increase in r results in an INR4.65 price decrease, underscoring the inverse relationship between price and YTM.

4. Now consider a bond issued at negative interest rates. In July 2016, Germany became the first eurozone country to issue 10-year sovereign bonds at a negative yield. If the German government bond annual YTM was −0.05 percent when issued, calculate the present value (PV) of the bond per EUR100 of principal (FV) at the time of issuance.

Solution:

EUR100.50
Solve for PV given r of −0.05 percent, t =10, and FV_{10} of EUR100 using Equation 1:

$$PV = \text{EUR}100.50 = \text{EUR}100 / (1 - 0.0005)^{10},$$

or

PV = (100.50) `= PV (-0.0005,10,0,100,0)`.

At issuance, this bond is priced at a **premium**, meaning that an investor purchasing the bond at issuance paid EUR0.50 above the future value expected at maturity, which is the principal.

5. Six years later, when German inflation reached highs not seen in decades and investors increased their required nominal rate of return, say we observed that the German government bond in question 4 is now trading at